Brownian Motion on the Unitary Group

MA Comprehensive Exam Mathematics

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Contents

1 Introduction

1.1 Brownian Motion on Non-Euclidean Spaces

On $\mathbb{R}^d$, Brownian motion is the diffusion generated by $\frac{1}{2}\Delta$, where Δ is the usual Euclidean Laplacian. Its transition probabilities solve the heat equation $\partial_t u = \frac{1}{2}\Delta u$, and the process itself can be described by putting independent Brownian motions in each coordinate direction. On a Riemannian manifold, the Euclidean Laplacian is replaced by the Laplace–Beltrami operator Δ_M: the corresponding diffusion is generated by $\frac{1}{2}\Delta_M$ and its transition probabilities solve the analogous heat equation on the manifold.

The goal of this thesis is threefold. First, we make this construction concrete and explicit in the important case of the unitary group $\mathrm{U}(n)$: we define Brownian motion on $\mathrm{U}(n)$ as the diffusion generated by $\frac{1}{2}\Delta$, where Δ is the Laplace–Beltrami operator for the bi-invariant Hilbert–Schmidt metric, and show that it admits four equivalent descriptions—via the martingale problem, the heat semigroup, the heat kernel, and an explicit Stratonovich SDE. Second, we develop the necessary background in Lie group geometry and representation theory, in particular the Peter–Weyl theorem, which plays the role of Fourier analysis on $\mathrm{U}(n)$. Third, we use this spectral theory to prove that the law of the process converges weakly to the normalized Haar measure as $t \to \infty$, establishing that Brownian motion on $\mathrm{U}(n)$ equidistributes over the group in the long run.

1.2 From Euclidean Space to the Unitary Group

On $\mathbb{R}^d$ one can simply put independent Brownian motions in each coordinate direction, and the resulting process moves freely through space. On $\mathrm{U}(n)$ this approach fails immediately. If one tries to run independent Brownian motions in the n^2 complex entries of a matrix, the constraint $U^*U = I$ is destroyed by the noise, the process leaves $\mathrm{U}(n)$. The constraint must be enforced at every instant, not just at the initial time. The resolution is that the noise must be tangent to $\mathrm{U}(n)$ at every point, meaning each infinitesimal step must lie in the tangent space $T_U\mathrm{U}(n)$. By Proposition 2.8, this tangent space equals $U\,\mathfrak{u}(n)$, the Lie algebra $\mathfrak{u}(n)$ of skew-Hermitian matrices transported to the point U by left multiplication. The Lie algebra is not merely an algebraic convenience; it is precisely the space through which Brownian noise must be injected into the group in order to preserve unitarity.

The second difficulty comes from noncommutativity. On the circle $\mathrm{U}(1)$, Brownian motion is generated by $\frac{1}{2}\frac{d^2}{d\theta^2}$. The Fourier modes $e^{ik\theta}$ are eigenfunctions satisfying

$$\frac{d^2}{d\theta^2}\, e^{ik\theta} = -k^2 e^{ik\theta},$$

so the heat semigroup sends $e^{ik\theta}$ to $e^{-k^2 t/2} e^{ik\theta}$. Every nonconstant mode decays exponentially, and only the constant $k = 0$ survives in the limit, corresponding to averaging against the uniform measure on the circle. For $n \geq 2$, $\mathrm{U}(n)$ is noncommutative, and classical Fourier series are no longer the right tool. The Peter–Weyl theorem provides the replacement: matrix coefficients of irreducible unitary representations take the role of Fourier modes, and the corresponding subspaces $\mathcal{H}_\pi$ replace the individual frequency components. The proof of convergence follows the same philosophy as the circle case, with the Peter–Weyl decomposition replacing Fourier series: every nontrivial mode has a strictly negative eigenvalue of Δ and decays under the heat semigroup, leaving only the constant mode in the limit.

Compactness of $\mathrm{U}(n)$ is essential throughout. On a compact group, Haar measure can be normalized to a probability measure, which makes it a natural candidate for the limiting distribution. Compactness also prevents the Brownian motion from escaping to infinity, so the process has no option but to spread and eventually fill the group uniformly. On a noncompact group such as $\mathrm{GL}_n(\mathbb{R})$, Haar measure is typically infinite, there is no candidate for a uniform equilibrium, the process may escape to infinity, and convergence to equilibrium is not automatic.

1.3 Main Result

The main goal of this thesis is to construct Brownian motion on $\mathrm{U}(n)$ as the diffusion with generator $\frac{1}{2}\Delta$, where Δ is the Laplace–Beltrami operator for the bi-invariant Hilbert–Schmidt metric, and to prove that its law ν_t converges weakly to the normalized Haar measure μ as $t \to \infty$. The same process admits four equivalent descriptions: as the solution to the martingale problem for $\frac{1}{2}\Delta$, via the heat semigroup P_t and the heat equation $\partial_t u = \frac{1}{2}\Delta u$, via the transition density given by the heat kernel $p_t(U, V)$, and as the explicit solution of a Stratonovich SDE on $\mathrm{U}(n)$.

1.4 Roadmap

Section 2 introduces $\mathrm{U}(n)$, its Lie algebra $\mathfrak{u}(n)$ of skew-Hermitian matrices, and the normalized Haar measure μ. The Lie algebra is the object that supplies the infinitesimal directions along which the Brownian noise is injected; Haar measure is the candidate limiting distribution toward which the process converges. Together they are the two central objects of the thesis.

Section 3 equips $\mathrm{U}(n)$ with a bi-invariant Riemannian metric, the Hilbert–Schmidt metric, and uses it to define the Laplace–Beltrami operator Δ. The bi-invariance is the key structural property: it forces Δ to commute with all left and right translations, which is the algebraic input that makes the spectral analysis in the final two sections possible.

Section 4 gives four equivalent constructions of Brownian motion on $\mathrm{U}(n)$ and shows that they all describe the same process. In particular, the Stratonovich SDE with coefficients in $\mathfrak{u}(n)$ is shown to remain in $\mathrm{U}(n)$ for all time, and the relationship between the stochastic and analytic descriptions is made explicit.

Section 5 develops the Peter–Weyl theorem, which decomposes $L^2(\mathrm{U}(n), \mu)$ into finite-dimensional irreducible blocks $\mathcal{H}_\pi$ and shows that Δ acts as a scalar multiple of the identity on each block. This is the noncommutative analogue of the Fourier decomposition on the circle, and it is the spectral tool that drives the convergence proof.

Section 6 applies the Peter–Weyl decomposition to prove the main theorem. Since each non-trivial block carries a strictly negative eigenvalue of Δ, the corresponding component of the heat semigroup decays exponentially, and only the constant mode survives in the limit, giving weak convergence $\nu_t \Rightarrow \mu$ as $t \to \infty$.

2 The Unitary Group

The unitary group $\mathrm{U}(n)$ is the group of all $n \times n$ complex matrices that preserve the standard Hermitian inner product.

2.1 Definition of $\mathrm{U}(n)$

Definition 2.1 (Unitary Group)**.** The *unitary group* is

$$\mathrm{U}(n) \;=\; \big\{U \in \mathbb{C}^{n\times n} : U^*U = I\big\}.$$

$\mathrm{U}(n)$ is a compact subgroup of $\mathrm{GL}_n(\mathbb{C})$: it is closed because $U^*U = I$ is a continuous condition, and bounded because its columns are unit vectors, so compactness follows by Heine–Borel. It is a subgroup because products and inverses of unitary matrices are unitary.

Example 2.2 ($\mathrm{U}(1)$: the unit circle)**.** For $n = 1$, the condition $U^*U = 1$ with $U \in \mathbb{C}$ reduces to $|U|^2 = 1$, so $\mathrm{U}(1) = \{e^{i\theta} : \theta \in \mathbb{R}\}$ is the unit circle in $\mathbb{C}$. Its Lie algebra is $\mathfrak{u}(1) = i\mathbb{R}$ (purely imaginary numbers, the 1×1 skew-Hermitian matrices).

2.2 The Lie Algebra $\mathfrak{u}(n)$

The Lie algebra is the linearized version of $\mathrm{U}(n)$ near the identity. It describes the infinitesimal directions in which one can move while remaining inside the group, and this is why it later becomes the natural space in which to place the Brownian noise.

Definition 2.3 (Lie Group)**.** A *Lie group* is a group that is also a finite-dimensional smooth manifold where the group operations (multiplication and inversion) are smooth maps.

Definition 2.4 (Lie Algebra)**.** The *Lie algebra* $\mathfrak{g}$ of a Lie group G is the tangent space T_eG at the identity, equipped with the Lie bracket $[\,\cdot\,,\,\cdot\,]$.

Proposition 2.5. *The Lie algebra of* $\mathrm{U}(n)$ *is*

$$\mathfrak{u}(n) \;=\; \big\{A \in \mathbb{C}^{n\times n} : A^* + A = 0\big\},$$

the space of skew-Hermitian matrices.

Proof. Recall that $\mathfrak{u}(n) = T_I\mathrm{U}(n)$ is the tangent space at the identity. We compute it directly by differentiating the defining constraint of $\mathrm{U}(n)$.

($\subseteq$) Let $U : (-\varepsilon, \varepsilon) \to \mathrm{U}(n)$ be a smooth curve with $U(0) = I$, and set $A := U'(0) \in T_I\mathrm{U}(n)$. Since $U(t) \in \mathrm{U}(n)$ for all t, we have

$$U(t)^*U(t) \;=\; I \qquad \text{for all } t.$$

Differentiating both sides with respect to t and evaluating at $t = 0$:

$$\frac{d}{dt}\big[U(t)^*U(t)\big]\Big|_{t=0} = U'(0)^*U(0) + U(0)^*U'(0)$$
$$= A^* + A \; = \; 0.$$

Hence every $A \in T_I\mathrm{U}(n)$ satisfies $A^* + A = 0$, i.e. A is skew-Hermitian.

($\supseteq$) Conversely, let $A \in \mathbb{C}^{n\times n}$ with $A^* + A = 0$. Define the curve $U(t) := e^{tA}$. Then $U(0) = I$ and $U'(0) = A$, so it remains to check $U(t) \in \mathrm{U}(n)$ for all t. Using $A^* = -A$:

$$U(t)^*U(t) = \big(e^{tA}\big)^* e^{tA} = e^{tA^*}e^{tA} = e^{-tA}e^{tA} = I.$$

Thus $e^{tA} \in \mathrm{U}(n)$ for all t, so $A = U'(0) \in T_I\mathrm{U}(n)$.

Together the two inclusions give $T_I\mathrm{U}(n) = \{A \in \mathbb{C}^{n\times n} : A^* + A = 0\}$. □

Remark 2.6. Let E_{ij} denote the matrix with a 1 in the (i, j)-entry and 0 elsewhere. A basis for $\mathfrak{u}(n)$ consists of: iE_{jj} for $1 \le j \le n$, and $E_{jk} - E_{kj}$, $i(E_{jk} + E_{kj})$ for $j < k$, giving $\dim \mathfrak{u}(n) = n^2$.

Remark 2.7. The condition $A^* + A = 0$ is the infinitesimal form of $U^*U = I$. To see why, substitute the first-order approximation $U(t) \approx I + tA$ into the unitarity constraint:

$$(I + tA)^*(I + tA) = I + t(A^* + A) + t^2A^*A.$$

The last term is $O(t^2)$ and vanishes in the limit $t \to 0$, so the condition that this expression returns to I at first order is exactly $A^* + A = 0$. In other words, $\mathfrak{u}(n)$ consists of those matrices for which $I + tA$ fails to be unitary only at second order, which is why injecting Brownian noise through $\mathfrak{u}(n)$ is the infinitesimal version of staying on the group.

2.3 Tangent Spaces and Translations

Proposition 2.8. *For each $U \in \mathrm{U}(n)$,*

$$T_U\mathrm{U}(n) \; = \; U\,\mathfrak{u}(n) \; = \; \mathfrak{u}(n)\,U.$$

Proof. Define the left translation map $L_U : \mathrm{U}(n) \to \mathrm{U}(n)$ by $L_U(V) = UV$. Since L_U is a diffeomorphism (its inverse is $L_{U^{-1}}$), its derivative at the identity is a linear isomorphism

$$dL_U\big|_I : T_I\mathrm{U}(n) = \mathfrak{u}(n) \;\xrightarrow{\sim}\; T_U\mathrm{U}(n).$$

We compute it explicitly: for a curve $V(t)$ in $\mathrm{U}(n)$ with $V(0) = I$ and $V'(0) = A \in \mathfrak{u}(n)$,

$$\frac{d}{dt}\big[U \cdot V(t)\big]\Big|_{t=0} = U \cdot V'(0) = UA.$$

Hence $dL_U|_I(A) = UA$, so $T_U\mathrm{U}(n) = U\,\mathfrak{u}(n)$. The identity $T_U\mathrm{U}(n) = \mathfrak{u}(n)\,U$ follows by the same argument applied to right translation $R_U(V) = VU$. □

2.4 Exponential Map and One-Parameter Subgroups

Proposition 2.9. *If $A \in \mathfrak{u}(n)$, then $e^{tA} \in \mathrm{U}(n)$ for all $t \in \mathbb{R}$, and the map $t \mapsto e^{tA}$ is a one-parameter subgroup of* $\mathrm{U}(n)$.

Proof. Unitarity. Using $A^* = -A$,

$$\left(e^{tA}\right)^* e^{tA} = e^{tA^*} e^{tA} = e^{-tA} e^{tA} = I,$$

so $e^{tA} \in \mathrm{U}(n)$ for all $t \in \mathbb{R}$.

One-parameter subgroup. Define $\varphi : \mathbb{R} \to \mathrm{U}(n)$ by $\varphi(t) = e^{tA}$. For any $s, t \in \mathbb{R}$, since sA and tA commute,

$$\varphi(s+t) = e^{(s+t)A} = e^{sA} e^{tA} = \varphi(s)\,\varphi(t).$$ □

2.5 Haar Measure

To state what it means for Brownian motion on $\mathrm{U}(n)$ to converge as $t \to \infty$, we need a canonical probability measure to converge to. Haar measure is the unique measure on $\mathrm{U}(n)$ that is invariant under the group action; it is, in a precise sense, the uniform distribution on $\mathrm{U}(n)$. Its existence follows from the general theory of locally compact groups.

For the construction, we first recall a representation theorem from measure theory.

Theorem 2.10 (Riesz-Markov-Kakutani Representation Theorem [Esp23])**.** *Let X be a locally compact Hausdorff space and ϕ a positive linear functional on $C_C(X)$. Then there exists a unique positive Borel measure μ on X such that*

$$\phi(f) = \int_X f(x)\,d\mu(x), \quad \forall f \in C_C(X)$$

Remark 2.11. We always normalize Haar measure so that $\int_G d\mu = 1$, making it a probability measure. For the rest of this paper, μ denotes the normalized Haar measure on $\mathrm{U}(n)$.

Theorem 2.12 (Haar [Tao11])**.** *On every locally compact (Hausdorff) topological group G, there exists a unique up to a constant left-invariant Borel-measure μ [Tao11].*

Proof. **Strategy.** We construct a left-invariant positive linear functional I on $C_c(G)$ via covering numbers, verify it is uniformly bounded, extract a convergent subnet using Tychonoff's theorem [Tao11], show the limit is additive, and then apply the Riesz–Markov–Kakutani theorem to produce the measure.

We define the left translation on G to be

$$\begin{aligned} L_g : G &\to G \\ h &\mapsto gh \end{aligned}$$

Which when applied to functions,

$$L_g f(x) := f(g^{-1}x)$$

In addition, by the Riesz-Markov-Kakutani Representation Theorem, we can construct a linear functional on the space of continuous functions with compact support, and show that the measure associated with it satisfies the definition of a Haar measure. Therefore, it suffices to show that there exists a positive linear functional, we will denote as I, such that it satisfies the properties of the Haar measure.

$$I : C_C(G)^+ \to \mathbb{R}^+$$

For $f \in C_C(G)^+$, we define the covering number as

$$[f : \varphi] = \inf \left\{ \sum_{i=1}^{n} c_i \colon c_i \in \mathbb{R}^+ f \leq \sum_{i=1}^{n} c_i L_{g_i} \varphi \; \forall g_i \in G \right\}$$

Fix a non-zero $f_0 \in C_C(G)^+$. Then define the normalised functional,

$$I_\varphi(f) := \frac{[f : \varphi]}{[f_0 : \varphi]}$$

We verify that I_φ satisfies the following properties:

1. (**Homogeneity**) Let $\lambda \in \mathbb{R}^+$. Consider $[\lambda f : \varphi]$

$$\begin{aligned} [\lambda f : \varphi] &= \inf \left\{ \sum_{i=1}^{n} c_i \colon \lambda f \leq \sum_{i=1}^{n} c_i L_{g_i} \varphi \right\} \\ &= \lambda \inf \left\{ \sum_{i=1}^{n} c_i' : f \leq \sum_{i=1}^{n} c_i' L_{g_i} \varphi \right\} \quad \text{(substituting } c_i' = c_i/\lambda) \\ &= \lambda \, [f : \varphi] \end{aligned}$$

Which when you divide by $[f_0 : \varphi]$ we get the required result.

2. (**Left-Invariance**) Let $h \in G$. Consider $[L_h f : \varphi]$

$$\begin{aligned}
[L_h f : \varphi] &= \inf \left\{ \sum_{i=1}^{n} c_i : L_h f \leq \sum_{i=1}^{n} c_i L_{g_i} \varphi \right\} \\
&= \inf \left\{ \sum_{i=1}^{n} c_i : f \leq \sum_{i=1}^{n} c_i L_{h^{-1} g_i} \varphi \right\} \quad \text{(applying } L_{h^{-1}} \text{ to both sides)} \\
&= \inf \left\{ \sum_{i=1}^{n} c_i : f \leq \sum_{i=1}^{n} c_i L_{g'_i} \varphi \right\} \quad \text{(setting } g'_i = h^{-1} g_i \text{, which ranges over all of } G\text{)} \\
&= [f : \varphi]
\end{aligned}$$

Dividing both sides by $[f_0 : \varphi]$ gives $I_\varphi(L_h f) = I_\varphi(f)$.

3. (**Monotonicity**) Let $f_1, f_2 \in C_C(G)^+$ with $f_1 \leq f_2$. Consider $[f_1 : \varphi]$

$$[f_1 : \varphi] = \inf \left\{ \sum_{i=1}^{n} c_i : f_1 \leq \sum_{i=1}^{n} c_i L_{g_i} \varphi \right\}$$

Since $f_1 \leq f_2$, any cover $\{c_i, g_i\}$ satisfying $f_2 \leq \sum_i c_i L_{g_i} \varphi$ also satisfies $f_1 \leq \sum_i c_i L_{g_i} \varphi$. Hence the feasible set for $[f_1 : \varphi]$ contains that of $[f_2 : \varphi]$, so

$$[f_1 : \varphi] \ \leq \ [f_2 : \varphi].$$

Dividing both sides by $[f_0 : \varphi]$ gives $I_\varphi(f_1) \leq I_\varphi(f_2)$.

4. (**Subadditivity**) Let $f_1, f_2 \in C_C(G)^+$. Let $\varepsilon > 0$ and choose near-optimal covers $\{c_i, g_i\}$ and $\{d_j, h_j\}$ such that

$$f_1 \ \leq \ \sum_{i=1}^{m} c_i L_{g_i} \varphi, \quad \sum_{i=1}^{m} c_i \ \leq \ [f_1 : \varphi] + \varepsilon,$$

$$f_2 \ \leq \ \sum_{j=1}^{k} d_j L_{h_j} \varphi, \quad \sum_{j=1}^{k} d_j \ \leq \ [f_2 : \varphi] + \varepsilon.$$

Adding these inequalities gives

$$f_1 + f_2 \ \leq \ \sum_{i=1}^{m} c_i L_{g_i} \varphi + \sum_{j=1}^{k} d_j L_{h_j} \varphi,$$

so the concatenated collection $\{c_i\} \cup \{d_j\}$ is a valid cover of $f_1 + f_2$, giving

$$[f_1 + f_2 : \varphi] \ \leq \ \sum_i c_i + \sum_j d_j \ \leq \ [f_1 : \varphi] + [f_2 : \varphi] + 2\varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, $[f_1 + f_2 : \varphi] \le [f_1 : \varphi] + [f_2 : \varphi]$. Dividing by $[f_0 : \varphi]$ gives $I_\varphi(f_1 + f_2) \le I_\varphi(f_1) + I_\varphi(f_2)$.

5. (**Uniform Bound**) We claim that for all nonzero $f, \varphi \in C_C(G)^+$,

$$\frac{1}{[f_0 : f]} \;\le\; I_\varphi(f) \;\le\; [f : f_0],$$

with both bounds independent of φ.

Upper bound. Any cover of f_0 by translates of φ can be chained with any cover of f by translates of f_0: if $f \le \sum_i c_i L_{g_i} f_0$ and $f_0 \le \sum_j d_j L_{h_j} \varphi$, then substituting gives

$$f \;\le\; \sum_i c_i L_{g_i} \left(\sum_j d_j L_{h_j} \varphi \right) = \sum_{i,j} c_i d_j \, L_{g_i h_j} \varphi.$$

Taking infima over both covers gives $[f : \varphi] \le [f : f_0][f_0 : \varphi]$, so

$$I_\varphi(f) = \frac{[f : \varphi]}{[f_0 : \varphi]} \;\le\; [f : f_0].$$

Lower bound. By the same transitivity argument with f and f_0 swapped, $[f_0 : \varphi] \le [f_0 : f][f : \varphi]$, hence

$$I_\varphi(f) = \frac{[f : \varphi]}{[f_0 : \varphi]} \;\ge\; \frac{1}{[f_0 : f]}.$$

6. (**Normalisation**) What is $I_\varphi(f_0)$? By definition,

$$I_\varphi(f_0) = \frac{[f_0 : \varphi]}{[f_0 : \varphi]} = 1.$$

So I_φ is already normalised with respect to f_0: regardless of the choice of φ, we always have $I_\varphi(f_0) = 1$. This is the scale-fixing condition that makes the uniform bound in property (5) meaningful and ensures the limiting functional $I := \lim_n I_{\varphi_n}$ is non-trivial.

Tychonoff extracts a convergent subnet. By property (5), for each fixed $f \in C_c(G)^+$ the value $I_\varphi(f)$ lies in the compact interval $[0, [f : f_0]]$. Form the product space

$$\mathcal{K} \;:=\; \prod_{f \in C_c(G)^+} \big[0,\, [f : f_0]\big].$$

By Tychonoff's theorem $\mathcal{K}$ is compact, and the net (I_φ) indexed by $\varphi \in C_c(G)^+$ directed by shrinking support toward e lies entirely in $\mathcal{K}$. Compactness provides a convergent subnet with limit

I. Convergence in the product topology is pointwise, so $I_{\varphi_\alpha}(f) \to I(f)$ for each fixed f. Since properties (1)–(4) are preserved under pointwise limits, I is left-invariant, positively homogeneous, monotone, and subadditive.

Approximate additivity becomes exact in the limit. For any $f_1, f_2 \in C_c(G)^+$ and $\varepsilon > 0$, once φ is supported on a sufficiently small neighbourhood of e, the splitting argument on the covering number gives

$$\left|I_\varphi(f_1 + f_2) - I_\varphi(f_1) - I_\varphi(f_2)\right| \;\le\; \varepsilon.$$

Passing to the limit along the convergent subnet and using pointwise convergence:

$$\left|I(f_1 + f_2) - I(f_1) - I(f_2)\right| \;\le\; \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, $I(f_1 + f_2) = I(f_1) + I(f_2)$. Combined with positive homogeneity, I extends to a positive linear functional on $C_c(G)$.

Apply Riesz-Markov-Kakutani to get the measure. By the Riesz-Markov-Kakutani there is a unique Radon measure μ on G with

$$I(f) \;=\; \int_G f\, d\mu \qquad \text{for all } f \in C_c(G).$$

Left-invariance of I gives $\mu(gA) = \mu(A)$ for every Borel $A \subseteq G$ and $g \in G$, so μ is a left-invariant Haar measure on G. □

Remark 2.13. The proof above establishes existence. Uniqueness up to a positive scalar is not proved here; it follows from the general theory of Haar measure on locally compact groups, see [Tao11].

Remark 2.14. The property that will be used throughout this thesis is left-invariance: for any $g \in \mathrm{U}(n)$ and continuous f,

$$\int_{\mathrm{U}(n)} f(gU)\, \mu(dU) = \int_{\mathrm{U}(n)} f(U)\, \mu(dU).$$

This makes μ the natural limiting distribution, since any stationary distribution of Brownian motion on $\mathrm{U}(n)$ must be left-invariant, and μ is the unique probability measure with this property.

3 Geometry of Lie Groups and $U(n)$

To define Brownian motion intrinsically on $\mathrm{U}(n)$, we need a notion of length, orthonormal directions, and a Laplacian. These are provided by a Riemannian metric, which assigns an inner product to each tangent space in a smoothly varying way.

On a general Riemannian manifold there is no canonical choice of metric, but $\mathrm{U}(n)$ is a Lie group and this extra structure singles one out. A metric that is invariant under all left and right translations, called bi-invariant, is the natural choice: it treats every point of the group the same way, and the resulting Laplacian inherits the symmetry of the group. This symmetry is what will later force Δ_n to commute with left and right translations, the key input for the Peter–Weyl analysis in Section 5.

The construction is simpler than it might appear. Because left translation is a diffeomorphism, a left-invariant metric is completely determined by what it does at a single point, namely the identity. So specifying the metric reduces to choosing an inner product on the Lie algebra $\mathfrak{u}(n)$, and then checking that it is Ad-invariant so that the metric is also right-invariant.

Proposition 3.1. *A left-invariant metric is completely determined by what it does at the identity.*

Proof. Let $\langle , \rangle$ be an inner product on the Lie algebra $\mathfrak{g} = T_e G$. Using the left translation map

$$L_g : G \to G \quad h \mapsto gh$$

Its derivative at the identity yields a linear isomorphism between tangent spaces:

$$dL_g|_e : T_e G = \mathfrak{g} \to T_g G$$

So if we know an inner product on $\mathfrak{g}$, we can push it to $T_g G$ for any g by declaring

$$\langle u, v \rangle := \left\langle (dL_g|_e)^{-1} u, (dL_g|_e)^{-1} v \right\rangle_e$$

□

However, in this case, it is necessary to equip our Lie group with a bi-invariant metric [Hal15].

Proposition 3.2. *A left-invariant Riemannian metric on a Lie group is right-invariant if and only if the corresponding inner product on the Lie algebra is Ad-invariant.*

Proof. Recall that the adjoint action is the action of the Lie group on its Lie algebra by conjugation

$$\mathrm{Ad}_U : \mathfrak{g} \to \mathfrak{g}, \quad A \mapsto UAU^{-1}$$

Let us consider what right translation does to the metric. Right translation by g is

$$R_g : h \mapsto hg$$

with derivative

$$dR_g|_h : T_h G \to T_{hg} G$$

Therefore, for the metric to be right invariant, we need that

$$\langle u, v\rangle_{hg} = \langle dR_g|_h u, dR_g|_h v\rangle_{hg}$$

A tangent vector $v \in T_hG$ corresponds to $(dL_h)^{-1}v \in \mathfrak{g}$. Under right translation by g, this vector gets sent to $T_{hg}G$, and pulling that back to $\mathfrak{g}$ via $(dL_{hg})^{-1}$ gives:

$$dL_{hg}^{-1} \circ dR_g|_h = d(L_h g^{-1} \circ R_g \circ L_h)|_e = d(Ad_g)|_e$$

Therefore, right-invariance of the metric reduces to

$$\langle Ad_g A, Ad_g B\rangle = \langle A, B\rangle$$

□

To define the Laplace-Beltrami operator on $\mathrm{U}(n)$, we need a Riemannian metric. By Proposition 3.1, it suffices to specify an Ad-invariant inner product on $\mathfrak{u}(n)$; the natural choice is the following normalized inner product.

Definition 3.3 (Inner Product on $\mathfrak{u}(n)$)**.** The inner product used throughout this thesis is

$$\langle A, B\rangle_n \;=\; -n\,\mathrm{Tr}(AB), \qquad A, B \in \mathfrak{u}(n).$$

This is n times the standard Hilbert–Schmidt inner product $-\mathrm{Tr}(AB)$. When $\{X_1, \ldots, X_{n^2}\}$ is called an orthonormal basis of $\mathfrak{u}(n)$, it is always orthonormal with respect to $\langle\cdot,\cdot\rangle_n$.

Lemma 3.4. *The inner product $\langle\cdot,\cdot\rangle_n$ of Definition 3.3 is a genuine inner product on $\mathfrak{u}(n)$.*

Proof. We check the three axioms. Throughout, $A, B \in \mathfrak{u}(n)$, so $A^* = -A$.

Symmetry. By cyclicity of the trace,

$$\langle A, B\rangle_n = -n\,\mathrm{Tr}(AB) = -n\,\mathrm{Tr}(BA) = \langle B, A\rangle_n.$$

Bilinearity. Follows immediately from linearity of the trace.

Positive definiteness. Using $A^* = -A$,

$$\langle A, A\rangle_n = -n\,\mathrm{Tr}(A^2) = n\,\mathrm{Tr}(A^*A) = n\sum_{i,j}|A_{ij}|^2 \;\geq\; 0,$$

with equality if and only if every entry of A is zero. □

Proposition 3.5. *The Hilbert-Schmidt inner product on $\mathfrak{u}(n)$ extends to a bi-invariant Riemannian metric on $\mathrm{U}(n)$.*

Proof. We already have a left-invariant metric: extend $\langle \cdot, \cdot \rangle$ from $\mathfrak{u}(n)$ to all of $\mathrm{U}(n)$ by left translation, as in Proposition 3.1.

It remains to check right-invariance. By Proposition 3.2, right-invariance of the metric is equivalent to Ad-invariance of the inner product on $\mathfrak{u}(n)$. For $U \in \mathrm{U}(n)$ and $A, B \in \mathfrak{u}(n)$,

$$\langle \mathrm{Ad}_U A,\, \mathrm{Ad}_U B \rangle_n = -n \operatorname{Tr}(UAU^{-1} \cdot UBU^{-1}) = -n \operatorname{Tr}(UABU^{-1}) = -n \operatorname{Tr}(AB) = \langle A, B \rangle_n,$$

where we used cyclicity of the trace in the last step. Hence the metric is bi-invariant. □

Proposition 3.6. *The Laplace–Beltrami operator on* $\mathrm{U}(n)$ *with the bi-invariant metric is*

$$\Delta_n f \;=\; \sum_{k=1}^{n^2} R_k^2 f,$$

where $\{X_k\}$ *is any orthonormal basis of* $\mathfrak{u}(n)$ *for* $\langle \cdot, \cdot \rangle_n$ *and* R_k *is the corresponding right-invariant vector field on* $\mathrm{U}(n)$*. With this normalization,* $\Delta_n = \frac{1}{n} \Delta_{\mathrm{HS}}$*, where* Δ_{HS} *is the Laplacian for the standard Hilbert–Schmidt metric* $-\operatorname{Tr}(AB)$*.*

Remark 3.7. Left- and right-invariant vector fields do not coincide in general; they agree only when the group is abelian. What is true is that because the metric on $\mathrm{U}(n)$ is bi-invariant, the Laplace–Beltrami operator can be written using either a left-invariant or a right-invariant orthonormal frame and gives the same result. This thesis uses the convention of right-invariant vector fields, so $\Delta f = \sum_{k=1}^{n^2} R_k^2 f$. Note also that Δ is nonpositive on nonconstant functions, as on the circle where $\frac{d^2}{d\theta^2} e^{ik\theta} = -k^2 e^{ik\theta}$.

4 Brownian Motion on $\mathrm{U}(n)$

The same $\mathrm{U}(n)$-valued diffusion admits four equivalent descriptions, each illuminating a different aspect of the same object, and the Main Theorem (Theorem 4.8) states their equivalence precisely. The first characterizes the process analytically via its *generator* $\frac{1}{2}\Delta$: for every smooth f, the process $f(U_t)$ minus the accumulated drift $\int_0^t \frac{1}{2} \Delta f(U_s)\, ds$ is a martingale. The second is the *heat semigroup* $P_t = e^{t\frac{1}{2}\Delta}$ and the *heat equation* $\partial_t u = \frac{1}{2}\Delta u$ it satisfies, which together encode how the distribution of the process evolves in time. The third is the *heat kernel* $p_t(U, V)$, the smooth transition density of the process with respect to Haar measure. The fourth is an explicit stochastic realization as the solution to an *Itô SDE* on $\mathrm{U}(n)$, derived from a Stratonovich equation whose coefficients lie in the Lie algebra $\mathfrak{u}(n)$ and therefore keep the process on the group. The diagram below displays how these descriptions fit together.

$$
\begin{tikzcd}
 & \text{Generator } \frac{1}{2}\Delta \arrow[d, "\text{martingale problem}"] & \\
 & \text{Brownian motion } U_t \text{ on } \mathrm{U}(n) \arrow[dl] \arrow[dr] & \\
\text{Semigroup } P_t \arrow[d] & & \text{Stratonovich SDE} \arrow[d] \\
\text{Heat equation } \partial_t u = \frac{1}{2}\Delta u \arrow[d] & & \text{Itô SDE} \\
\text{Transition kernel } p_t(U, dV) \arrow[d] & & \\
\text{Heat kernel } p_t(U, V) & &
\end{tikzcd}
$$

4.1 Brownian Motion via the Martingale Problem

Rather than constructing the process via an SDE, we first characterize it analytically: a process is Brownian motion on $\mathrm{U}(n)$ if and only if it turns every smooth function into a martingale after subtracting the drift prescribed by the generator $\frac{1}{2}\Delta$. This *martingale problem* formulation is well suited to proving uniqueness and to connecting the process to the heat equation.

We start from the generator $\frac{1}{2}\Delta$ and define Brownian motion as the solution to the corresponding martingale problem. To state this precisely: a process (U_t) has generator $\frac{1}{2}\Delta$ if, for every smooth test function f, the deterministic drift of $f(U_t)$ is $\frac{1}{2}\Delta f(U_t)$ at each instant, and after subtracting this accumulated drift the remaining fluctuation is a martingale.

Definition 4.1 (Brownian Motion on $\mathrm{U}(n)$)**.** We say $(U_t)_{t\geq 0}$ is a *Brownian motion on* $\mathrm{U}(n)$ if for every $f \in C^\infty(\mathrm{U}(n))$, the process

$$M_t^f \;:=\; f(U_t) - f(U_0) - \int_0^t \tfrac{1}{2}\Delta f(U_s)\, ds$$

is a martingale with respect to the natural filtration $\mathcal{F}_t$.

Since $\frac{1}{2}\Delta$ is a second-order elliptic operator on the compact manifold $\mathrm{U}(n)$, the martingale problem is well-posed: for every starting point $U_0 \in \mathrm{U}(n)$ there exists a unique-in-law solution $(U_t)_{t\geq 0}$ [Hsu02, Theorems 1.3.4 and 1.3.6].

Remark 4.2. One can see that the Laplace-Beltrami operator is elliptic through the realization that $T_U(U(n)) = u(n)U$ and choosing an orthonormal basis X_k. Then the span of R_k is the span of the tangent spaces at $U(n)$ which is the required result.

4.2 Semigroup, Heat Equation, and Heat Kernel

The heat semigroup. For $f \in C^\infty(\mathrm{U}(n))$ and $U \in \mathrm{U}(n)$, define

$$P_t f(U) \;:=\; \mathbb{E}^U\big[f(U_t)\big] = \mathbb{E}\left[f(U_t) \mid U_0 = U\right].$$

Proposition 4.3. *The family $(P_t)_{t\geq 0}$ is a semigroup on $C(\mathrm{U}(n))$, meaning it satisfies:*

1. (Identity) $P_0 = \mathrm{id}$.
2. (Semigroup law) $P_{s+t} = P_s \circ P_t$ *for all* $s, t \geq 0$.

Proof. Throughout, $f \in C(\mathrm{U}(n))$, $U \in \mathrm{U}(n)$, and $(U_t)_{t\geq 0}$ denotes Brownian motion started at $U_0 = U$.

(1) Identity. At $t = 0$ the process has not moved, so

$$P_0 f(U) = \mathbb{E}^U[f(U_0)] = f(U).$$

(2) Semigroup law. By the tower property of conditional expectation, conditioning on $\mathcal{F}_s$:

$$P_{s+t} f(U) = \mathbb{E}^U[f(U_{s+t})] = \mathbb{E}^U\big[\mathbb{E}^U[f(U_{s+t}) \mid \mathcal{F}_s]\big].$$

The inner conditional expectation depends on the future evolution after time s. By the Markov property of (U_t), this future depends only on the current position U_s:

$$\mathbb{E}^U[f(U_{s+t}) \mid \mathcal{F}_s] = \mathbb{E}^{U_s}[f(U_t)] = P_t f(U_s).$$

Substituting back,

$$P_{s+t} f(U) = \mathbb{E}^U[P_t f(U_s)] = P_s(P_t f)(U) = (P_s \circ P_t) f(U).$$

$\square$

4.3 From the Semigroup to the Heat Equation

The name *heat semigroup* is justified by the following derivation. Since M_t^f is a martingale, its expectation is constant, so $\mathbb{E}^U[M_t^f] = 0$ for all $t \geq 0$. Expanding the definition of M_t^f and rearranging,

$$\mathbb{E}^U[f(U_t)] - f(U) \;=\; \int_0^t \mathbb{E}^U\left[\tfrac{1}{2}\Delta f(U_s)\right] ds \;=\; \int_0^t P_s\big(\tfrac{1}{2}\Delta f\big)(U)\, ds,$$

where in the last step we used the fact that $\mathbb{E}^U[\frac{1}{2}\Delta f(U_s)]$ is, by definition of the semigroup, equal to $P_s(\frac{1}{2}\Delta f)(U)$. Recognizing the left-hand side as $P_t f(U) - P_0 f(U)$ and differentiating both sides with respect to t via the Fundamental Theorem of Calculus gives $\partial_t P_t f(U) = P_t(\frac{1}{2}\Delta f)(U)$. Using the smoothing regularity of P_t on the compact manifold $\mathrm{U}(n)$ together with the semigroup property, P_t and Δ commute, so this becomes

$$\partial_t P_t f \;=\; \tfrac{1}{2}\Delta P_t f.$$

Setting $u(t,U) := P_t f(U)$ we recover the heat equation $\partial_t u = \frac{1}{2}\Delta u$, which explains why $(P_t)_{t\geq 0}$ is called the *heat semigroup*.

Transition probabilities. Fix $t > 0$ and $U \in \mathrm{U}(n)$. The map $f \mapsto P_t f(U)$ is linear, positive, and satisfies $P_t\mathbf{1}(U) = 1$. By the Riesz–Markov–Kakutani theorem (Theorem 2.10), there exists a unique probability measure $p_t(U, dV)$ on $\mathrm{U}(n)$ such that

$$P_t f(U) \;=\; \int_{\mathrm{U}(n)} f(V)\, p_t(U, dV).$$

Remark 4.4. Since $p_t(U, dV)$ is a probability measure on $\mathrm{U}(n)$, by construction $U_t \in U(n)$.

The heat kernel. Since $\mathrm{U}(n)$ is compact and Δ is elliptic, the transition measure $p_t(U, dV)$ has a smooth density with respect to the normalized Haar measure μ [Lia04, p. 6]:

$$p_t(U, dV) \;=\; p_t(U, V)\, \mu(dV).$$

The smooth function $(t, U, V) \mapsto p_t(U, V)$ is the *heat kernel* on $\mathrm{U}(n)$. When $U_0 = I$, the law of U_t is $p_t(I, V)\, \mu(dV)$.

Remark 4.5. The generator $\frac{1}{2}\Delta$, the heat semigroup, the heat equation, the transition probabilities, and the heat kernel are all equivalent descriptions of the same diffusion on $\mathrm{U}(n)$.

4.4 Realization via a Stratonovich SDE

One might ask: why not simply write an Itô SDE with diffusion coefficients in $\mathfrak{u}(n)$ and call it done? The issue is that the Itô integral does not respect the constraint $U^*U = I$. If one writes $dU_t = \sum_k X_k U_t\, dB_t^k$ in the Itô sense (with no drift), the quadratic variation contributes an extra term to $d(U_t^* U_t)$ that does not cancel; the process immediately leaves $\mathrm{U}(n)$. This is visible in Proposition 4.7: the drift $-\frac{1}{2}U_t\, dt$ in the Itô SDE is precisely the correction needed to keep $d(U_t^* U_t) = 0$.

The Stratonovich integral, by contrast, satisfies the classical chain rule. A Stratonovich SDE with coefficients that are tangent to $\mathrm{U}(n)$ therefore stays in $\mathrm{U}(n)$ automatically, with no need for a correction drift. This is what makes Stratonovich the natural calculus for SDEs on Lie groups. By Proposition 2.8, the diffusion coefficients $X_k U_t$ belong to $T_{U_t}\mathrm{U}(n) = \mathfrak{u}(n)\, U_t$, so the noise is always pointing along the group. Unitarity is then a consequence of the chain rule, not a coincidence of the drift.

Stratonovich formulation. Let $\{X_1, \ldots, X_{n^2}\}$ be a basis of $\mathfrak{u}(n)$ that is orthonormal with respect to the inner product $\langle \cdot, \cdot \rangle_n = -n \operatorname{Tr}(\cdot\, \cdot)$ from Definition 3.3, and let $\{B_t^k\}_{k=1}^{n^2}$ be independent standard Brownian motions. Define U_t by the *Stratonovich SDE*

$$dU_t \;=\; \sum_{k=1}^{n^2} X_k U_t \circ dB_t^k.$$

Proposition 4.6. *The generator of the process defined by the Stratonovich SDE above is* $\frac{1}{2}\Delta$*.*

Proof. Define right-invariant vector fields $R_k f(U) := \mathcal{D}f(U)[X_k U]$. By the Stratonovich chain rule,

$$df(U_t) \;=\; \sum_{k=1}^{n^2} (R_k f)(U_t) \circ dB_t^k.$$

Converting to Itô form introduces the correction $\frac{1}{2}\sum_k d[(R_k f)(U_t),\, B_t^k] = \frac{1}{2}\sum_k R_k^2 f(U_t)\, dt$, giving

$$df(U_t) \;=\; \sum_{k=1}^{n^2} (R_k f)(U_t)\, dB_t^k \;+\; \frac{1}{2}\sum_{k=1}^{n^2} R_k^2 f(U_t)\, dt.$$

The dt coefficient is the generator, so $\mathcal{A}f = \frac{1}{2}\sum_k R_k^2 f = \frac{1}{2}\Delta f$ by Proposition 3.6. ☐

Conversion to Itô form. The Stratonovich-to-Itô correction yields

$$dU_t \;=\; \sum_{k=1}^{n^2} X_k U_t\, dB_t^k \;+\; \frac{1}{2}\sum_{k=1}^{n^2} X_k^2\, U_t\, dt.$$

Using the identity $\sum_{k=1}^{n^2} X_k^2 = -I$ (which follows from $X_k = iM_k$ with $\{M_k\}$ orthonormal and $\sum_k M_k^2 = I$),

$$dU_t \;=\; \sum_{k=1}^{n^2} X_k U_t\, dB_t^k \;-\; \frac{1}{2} U_t\, dt.$$

Set $i\, dH_t := \sum_{k=1}^{n^2} X_k\, dB_t^k$, so H_t takes values in Hermitian matrices. Because $\{X_k\}$ is orthonormal for $\langle \cdot, \cdot \rangle_n$, the quadratic variation satisfies $(dH_t)^2 = I\, dt$, which makes the unitarity calculation $d(U_t^* U_t) = 0$ transparent, as we now verify. With this notation the *Itô SDE* is

$$dU_t \;=\; i\, dH_t\, U_t \;-\; \frac{1}{2} U_t\, dt. \tag{1}$$

Proposition 4.7 (Unitarity)**.** *The solution U_t of* (1) *satisfies $d(U_t^* U_t) = 0$, so $U_t \in \mathrm{U}(n)$ for all $t \geq 0$.*

Proof. **Strategy.** Apply the Itô product rule to $U_t^* U_t$ and show all three resulting terms sum to zero by using the conjugate-transpose of the SDE and the quadratic variation of H_t.

By the Itô product rule, $d(U_t^* U_t) = U_t^* \, dU_t + dU_t^* \, U_t + dU_t^* \, dU_t$. Since H_t takes values in Hermitian matrices ($-iX_k$ is Hermitian for each k), taking conjugate transposes of (1) gives $dU_t^* = -i \, U_t^* \, dH_t - \frac{1}{2} U_t^* \, dt$. Computing term by term:

$$\begin{aligned} U_t^* \, dU_t &= \quad i \, U_t^* \, dH_t \, U_t - \tfrac{1}{2} I \, dt, \\ dU_t^* \, U_t &= -i \, U_t^* \, dH_t \, U_t - \tfrac{1}{2} I \, dt, \\ dU_t^* \, dU_t &= U_t^* (dH_t)^2 \, U_t. \end{aligned}$$

For the quadratic variation, using $i \, dH_t = \sum_k X_k \, dB_t^k$ so $dH_t = -i \sum_k X_k \, dB_t^k$:

$$(dH_t)^2 \;=\; (-i)^2 \sum_{k=1}^{n^2} X_k^2 \, dt \;=\; -(-I) \, dt \;=\; I \, dt.$$

Adding all three terms: $d(U_t^* U_t) = (-\frac{1}{2} - \frac{1}{2} + 1) \, I \, dt = 0$. ☐

4.5 Main Theorem

Theorem 4.8. *The following four objects determine the same $\mathrm{U}(n)$-valued diffusion:*

1. *The diffusion with generator $\frac{1}{2}\Delta$, characterized by the martingale problem (Definition 4.1).*
2. *The Markov process with heat semigroup $P_t = e^{t\frac{1}{2}\Delta}$, which solves the heat equation $\partial_t u = \frac{1}{2}\Delta u$.*
3. *The process with transition density given by the heat kernel $p_t(U, V)$ via $\mathcal{L}(U_t \mid U_0 = U)(dV) = p_t(U, V) \, \mu(dV)$.*
4. *The solution to the Itô SDE* (1)*: $dU_t = i \, dH_t \, U_t - \frac{1}{2} U_t \, dt$.*

5 The Peter–Weyl Theorem

To analyze the long-run behavior of Brownian motion on $\mathrm{U}(n)$, we need to understand the spectrum of the Laplacian Δ. The Peter–Weyl theorem gives a complete spectral decomposition of $L^2(G, \mu)$ in terms of representation theory, playing the same role that Fourier analysis plays on the circle.

Definition 5.1 (Unitary representation)**.** A *finite-dimensional unitary representation* of a compact group G is a group homomorphism $\pi : G \to \mathrm{U}(V_\pi)$, where V_π is a finite-dimensional complex Hilbert space. The representation is *irreducible* if the only G-invariant subspaces of V_π are $\{0\}$ and V_π.

Fix an irreducible unitary representation π and let $d_\pi = \dim V_\pi$. Choose an orthonormal basis $\{e_1, \dots, e_{d_\pi}\}$ of V_π and define the *matrix coefficient functions*

$$\pi_{ij}(g) \;:=\; \langle \pi(g)\, e_j,\, e_i \rangle, \qquad 1 \le i, j \le d_\pi.$$

On $U(1) = \{e^{i\theta}\}$, the irreducible representations are the maps $e^{i\theta} \mapsto e^{ik\theta}$ for $k \in \mathbb{Z}$, whose matrix coefficients are exactly the Fourier modes $e^{ik\theta}$. Every square-integrable function on the circle expands into these modes, and each mode is an eigenfunction of the Laplacian. The functions π_{ij} are the noncommutative analogues of those modes on a general compact group. They build up arbitrary functions in $L^2(G, \mu)$ and, as we will see, each block $\mathcal{H}_\pi$ is an eigenspace of Δ.

Set

$$\mathcal{H}_\pi \;:=\; \operatorname{span}\bigl\{\pi_{ij} : 1 \le i, j \le d_\pi\bigr\} \;\subseteq\; L^2(G, \mu).$$

The space $\mathcal{H}_\pi$ is the noncommutative counterpart of a single frequency block. On the circle, the span of $e^{ik\theta}$ for a fixed k is the frequency-k block; on a general compact group, $\mathcal{H}_\pi$ collects all matrix coefficients of π into a single block of the same kind. Different representations correspond to different frequencies, and the Peter–Weyl theorem asserts that these blocks are mutually orthogonal and together span all of $L^2(G, \mu)$.

Let $\widehat{G}$ denote the set of equivalence classes of irreducible unitary representations of G.

Theorem 5.2 (Peter–Weyl [Tao11, App14, Wil])**.** *Let G be a compact group. For each irreducible unitary representation π of G:*

- *Each $\mathcal{H}_\pi$ is finite-dimensional.*
- *If $\pi \ncong \sigma$, then $\mathcal{H}_\pi \perp \mathcal{H}_\sigma$ in $L^2(G, \mu)$.*
- *The family $\bigl\{\sqrt{d_\pi}\, \pi_{ij} : \pi \in \widehat{G},\ 1 \le i, j \le d_\pi\bigr\}$ forms an orthonormal basis of $L^2(G, \mu)$.*
- $L^2(G, \mu) \;=\; \bigoplus_{\pi \in \widehat{G}} \mathcal{H}_\pi \;\cong\; \bigoplus_{\pi \in \widehat{G}} V_\pi^* \otimes V_\pi.$
- *The linear span of $\bigl\{\pi_{ij} : \pi \in \widehat{G},\ 1 \le i, j \le d_\pi\bigr\}$ is uniformly dense in $C(G)$ [App14].*

The Peter–Weyl theorem is the noncommutative analogue of the classical Fourier theorem on the circle. Just as every square-integrable function on $[0, 2\pi)$ decomposes uniquely into Fourier modes $e^{ik\theta}$, every function in $L^2(G, \mu)$ decomposes uniquely into matrix coefficients drawn from the blocks $\mathcal{H}_\pi$. The orthogonality statement says different representations contribute independent frequency components, and the basis statement says these components together account for everything. The fifth point strengthens this. Not only do the matrix coefficients span $L^2(G, \mu)$ in the Hilbert space sense, their finite linear combinations are uniformly dense in $C(G)$. This means Peter–Weyl polynomials approximate continuous functions as well in the supremum norm as in the L^2 norm, a fact we will use in Section 6 to promote pointwise convergence of the heat semigroup on polynomials to weak convergence of the law of U_t.

5.1 Δ Preserves the Peter–Weyl Blocks

On the circle, $\frac{d^2}{d\theta^2}$ maps each frequency block to itself by acting as a scalar on every mode $e^{ik\theta}$. The same is true of Δ on G, and the reason is geometric. The metric on G is bi-invariant, so left multiplication $L_a(g) = ag$ and right multiplication $R_b(g) = gb$ are isometries for every $a, b \in G$. Since the Laplace–Beltrami operator is defined entirely by the metric, it commutes with every isometry of the manifold. Translating a function and then differentiating gives the same result as differentiating and then translating. The blocks $\mathcal{H}_\pi$ are themselves built out of these translational symmetries. As Lemma 5.4 below confirms, each is stable under all left and right translations. An operator that commutes with all such translations cannot move a function out of a subspace already stable under them, so Δ must map each $\mathcal{H}_\pi$ to itself.

For $a, b \in G$, define an action of $G \times G$ on functions by

$$\big((a, b) \cdot f\big)(g) \;:=\; f(a^{-1}gb).$$

The left variable acts by left translation and the right variable by right translation.

Proposition 5.3. *Because the metric on G is bi-invariant, every left and right translation is an isometry. Consequently, for every smooth f and every $a, b \in G$,*

$$\Delta(f \circ L_a) = (\Delta f) \circ L_a \qquad \textit{and} \qquad \Delta(f \circ R_b) = (\Delta f) \circ R_b.$$

In other words, Δ commutes with the $G \times G$-action on functions.

Proof. Since the metric is bi-invariant, left and right translations are isometries of the Riemannian manifold G. The Laplace–Beltrami operator is defined purely in terms of the metric, so it is equivariant under isometries: $\Delta(f \circ \phi) = (\Delta f) \circ \phi$ for any isometry ϕ. Applying this to $\phi = L_a$ and $\phi = R_b$ gives the result. □

Lemma 5.4. *For every irreducible unitary representation π, the space $\mathcal{H}_\pi$ is invariant under left and right translations.*

Proof. Let $a, g \in G$. Using the homomorphism property of π,

$$\begin{aligned}\pi_{ij}(a^{-1}g) &= \langle \pi(a^{-1}g)\, e_j,\, e_i\rangle = \langle \pi(a^{-1})\pi(g)\, e_j,\, e_i\rangle \\ &= \sum_{k=1}^{d_\pi} \pi(a^{-1})_{ik}\, \pi_{kj}(g) \;\in\; \mathcal{H}_\pi.\end{aligned}$$

Hence $\mathcal{H}_\pi$ is invariant under left translations. An analogous computation using $\pi_{ij}(gb) = \sum_k \pi_{ik}(g)\,\pi(b)_{kj}$ gives invariance under right translations. □

Corollary 5.5. *For every irreducible representation π, one has $\Delta(\mathcal{H}_\pi) \subseteq \mathcal{H}_\pi$.*

Proof. By Lemma 5.4, $\mathcal{H}_\pi$ is stable under the $G \times G$-action. Since Δ commutes with that action (Proposition 5.3), the conclusion follows. □

5.2 Δ is Scalar on Each Peter–Weyl Block

$\mathcal{H}_\pi$ is in general *not* irreducible for the left G-action alone. To see why, recall that the left G-action on $\mathcal{H}_\pi$ is $(a \cdot \pi_{ij})(g) = \pi_{ij}(a^{-1}g) = \sum_k \pi(a^{-1})_{ik}\,\pi_{kj}(g)$, so it mixes the first index i while leaving the second index j fixed. For each fixed j, the subspace

$$E_j \;:=\; \mathrm{span}\{\pi_{ij} : 1 \le i \le d_\pi\}$$

is therefore left-invariant. Moreover, the map $e_i \mapsto \pi_{ij}$ identifies E_j with V_π as a left G-module (since the left action on π_{ij} reproduces exactly the action of $\pi(a^{-1})$ on the e_i-basis). There are d_π such subspaces, one for each column index j, giving the decomposition

$$\mathcal{H}_\pi \;=\; \bigoplus_{j=1}^{d_\pi} E_j \;\cong\; V_\pi^{\oplus d_\pi}$$

as left G-modules. Since $d_\pi \ge 1$ and each $E_j \cong V_\pi$ is a proper subspace whenever $d_\pi > 1$, $\mathcal{H}_\pi$ is reducible as a left G-module: Schur's lemma would only force Δ to be scalar on each E_j separately, not on all of $\mathcal{H}_\pi$. The correct viewpoint is to use the full action of $G \times G$, under which $\mathcal{H}_\pi$ *is* irreducible, as we now show.

Theorem 5.6 (Schur's Lemma)**.** *Let W be a finite-dimensional irreducible complex representation of a group H. If $T : W \to W$ commutes with the action of H, then there exists $c \in \mathbb{C}$ such that $T = c\,\mathrm{Id}_W$.*

Proof. Since W is a finite-dimensional vector space over $\mathbb{C}$, the operator T has at least one eigenvalue $c \in \mathbb{C}$. The corresponding eigenspace

$$E_c \;:=\; \ker(T - c\,\mathrm{Id}_W) \;\neq\; \{0\}$$

is H-invariant: for any $w \in E_c$ and $h \in H$,

$$T\big(\pi(h)w\big) = \pi(h)\big(Tw\big) = \pi(h)(cw) = c\,\pi(h)w,$$

so $\pi(h)w \in E_c$. Since W is irreducible and $E_c \neq \{0\}$, we must have $E_c = W$, i.e. $T = c\,\mathrm{Id}_W$. □

Proposition 5.7. *As a representation of $G \times G$, one has $\mathcal{H}_\pi \cong V_\pi^* \otimes V_\pi$. This representation is irreducible. Under this identification, the left copy of G acts on V_π^* via the contragredient and the right copy on V_π, so the two indices of the matrix coefficients correspond to two separate irreducible directions.*

Proof. Since $\mathcal{H}_\pi \subseteq L^2(G, \mu)$ is a space of functions on G, it inherits the $G \times G$-action on functions defined in Proposition 5.3:

$$\big((a,b) \cdot f\big)(g) \;=\; f(a^{-1}gb).$$

We endow $V_\pi^* \otimes V_\pi$ with the natural $G \times G$-action: the left factor acts on V_π^* via the *contragredient* of π,

$$(a \cdot \phi)(w) \;:=\; \phi\big(\pi(a)^{-1}w\big), \qquad \phi \in V_\pi^*,\; w \in V_\pi,$$

and the right factor acts on V_π via π itself, so altogether

$$(a,b) \cdot (\phi \otimes v) \;:=\; (a \cdot \phi) \otimes \pi(b)v.$$

Step 1: Isomorphism. Define

$$\Phi : V_\pi^* \otimes V_\pi \;\longrightarrow\; \mathcal{H}_\pi, \qquad \phi \otimes v \;\longmapsto\; \big[g \mapsto \phi(\pi(g)v)\big].$$

On basis elements, $\Phi(e_i^* \otimes e_j)(g) = e_i^*(\pi(g)e_j) = \pi_{ij}(g)$, so Φ sends the basis $\{e_i^* \otimes e_j\}$ bijectively to $\{\pi_{ij}\}$. Hence Φ is a linear isomorphism.

Step 2: Φ is $G \times G$-equivariant. For any $(a,b) \in G \times G$ and $\phi \otimes v \in V_\pi^* \otimes V_\pi$,

$$\begin{aligned}
\Phi\big((a,b) \cdot (\phi \otimes v)\big)(g) &= \Phi\big((a \cdot \phi) \otimes \pi(b)v\big)(g) \\
&= (a \cdot \phi)\big(\pi(g)\pi(b)v\big) \\
&= \phi\big(\pi(a)^{-1}\pi(g)\pi(b)v\big) \\
&= \phi\big(\pi(a^{-1}gb)v\big) \\
&= \Phi(\phi \otimes v)(a^{-1}gb) \\
&= \big((a,b) \cdot \Phi(\phi \otimes v)\big)(g).
\end{aligned}$$

Hence $\Phi : V_\pi^* \otimes V_\pi \xrightarrow{\sim} \mathcal{H}_\pi$ is an isomorphism of $G \times G$-representations.

Step 3: The contragredient V_π^* is irreducible. Let $U \subseteq V_\pi^*$ be a G-invariant subspace under the contragredient action, and let $U^0 := \{v \in V_\pi : \phi(v) = 0 \;\forall \phi \in U\}$ be its annihilator. For any $v \in U^0$, $g \in G$, and $\phi \in U$:

$$\phi\big(\pi(g)v\big) = (g^{-1} \cdot \phi)(v) = 0,$$

since $g^{-1} \cdot \phi \in U$ (as U is G-invariant) and $v \in U^0$. So $\pi(g)v \in U^0$, making U^0 a G-invariant subspace of V_π. By irreducibility of V_π, either $U^0 = \{0\}$, forcing $U = V_\pi^*$, or $U^0 = V_\pi$, forcing $U = \{0\}$. Hence V_π^* is irreducible.

Step 4: $V_\pi^* \otimes V_\pi$ is irreducible. Let

$$T \in \mathrm{End}_{G\times G}(V_\pi^* \otimes V_\pi),$$

so T is a linear map commuting with the $G \times G$-action.

We first use equivariance under the right factor $(1, b)$, which acts as $\mathrm{Id}_{V_\pi^*} \otimes \pi(b)$. Since the second copy of G acts only on the V_π-factor, the map T must commute with $\pi(b)$ on that factor for every $b \in G$. Because V_π is irreducible, Schur's Lemma (Theorem 5.6) implies that any linear map commuting with $\pi(G)$ is scalar. Therefore T must be of the form

$$T = C \otimes \mathrm{Id}_{V_\pi}$$

for some linear map $C \in \mathrm{End}(V_\pi^*)$.

Next we use equivariance under the left factor $(a, 1)$, which acts only on V_π^* via the contragredient representation. Since T commutes with this action, we have

$$C(a \cdot \phi) = a \cdot C(\phi) \qquad \forall\, a \in G,\ \phi \in V_\pi^*.$$

Thus C is a G-equivariant endomorphism of V_π^*. Since V_π^* is irreducible by Step 3, Schur's Lemma again gives

$$C = c\,\mathrm{Id}_{V_\pi^*} \qquad \text{for some } c \in \mathbb{C}.$$

Hence

$$T = c\,\mathrm{Id}_{V_\pi^* \otimes V_\pi}.$$

So every $G \times G$-equivariant endomorphism of $V_\pi^* \otimes V_\pi$ is scalar.

To conclude irreducibility, suppose $0 \neq W \subseteq V_\pi^* \otimes V_\pi$ is a $G \times G$-invariant subspace. Since the representation is unitary and finite-dimensional, its orthogonal complement $W^\perp$ is also $G \times G$-invariant. Let $P : V_\pi^* \otimes V_\pi \to W$ be the orthogonal projection. Then P commutes with the $G \times G$-action, so by the above, $P = c\,\mathrm{Id}$ for some $c \in \mathbb{C}$. Since P is a projection, $P^2 = P$, hence $c^2 = c$, so $c = 0$ or $c = 1$. If $c = 0$ then $P = 0$, forcing $W = \{0\}$, a contradiction. Thus $c = 1$, so $P = \mathrm{Id}$ and $W = V_\pi^* \otimes V_\pi$. Hence there is no nonzero proper invariant subspace, and $V_\pi^* \otimes V_\pi$ is irreducible. □

Corollary 5.8. *For every irreducible representation π of G, there exists $\lambda_\pi \geq 0$ such that*

$$\Delta f \;=\; -\lambda_\pi\, f \qquad \text{for all } f \in \mathcal{H}_\pi.$$

This is the noncommutative analogue of the fact that $\frac{d^2}{d\theta^2}$ acts as the scalar $-k^2$ on the Fourier mode $e^{ik\theta}$. Each frequency block $\mathcal{H}_\pi$ turns out to be an eigenspace of Δ, with eigenvalue $-\lambda_\pi$ playing the role of $-k^2$. The following argument makes this precise through four logical steps.

Proof. By Proposition 5.3, Δ commutes with every left translation L_a and every right translation R_b, since the metric is bi-invariant and all translations are therefore isometries. Acting by $(a, b) \in G \times G$ on a function amounts to composing left translation by a^{-1} with right translation by b, so commutativity with each translation individually implies commutativity with the full $G \times G$-action. By Corollary 5.5, Δ maps $\mathcal{H}_\pi$ into itself and so restricts to a well-defined endomorphism $\Delta|_{\mathcal{H}_\pi} : \mathcal{H}_\pi \to \mathcal{H}_\pi$. That restricted operator inherits commutativity with the $G \times G$-action, since the commutativity holds on all of $L^2(G, \mu)$ and therefore on every invariant subspace. By Proposition 5.7, $\mathcal{H}_\pi \cong V_\pi^* \otimes V_\pi$ is irreducible as a $G \times G$-module, so Schur's Lemma (Theorem 5.6) forces $\Delta|_{\mathcal{H}_\pi} = c_\pi \, \mathrm{Id}_{\mathcal{H}_\pi}$ for some $c_\pi \in \mathbb{C}$. Self-adjointness of Δ makes c_π real, and non-positivity gives $c_\pi \leq 0$. Writing $\lambda_\pi = -c_\pi \geq 0$ completes the proof. $\square$

Proposition 5.9. *One has $\lambda_\pi = 0$ if and only if π is the trivial representation.*

Proof. By Corollary 5.8, $\lambda_\pi = 0$ if and only if every $f \in \mathcal{H}_\pi$ satisfies $\Delta f = 0$. It therefore suffices to show that the only harmonic functions on G are the constants, since the constants span exactly $\mathcal{H}_{\mathbf{1}}$, the Peter–Weyl block of the trivial representation.

Let $f \in C^\infty(G)$ satisfy $\Delta f = 0$. Multiplying both sides by f and integrating over G against the Haar measure μ gives

$$0 \;=\; \int_G f \, \Delta f \, d\mu.$$

An integration by parts on the manifold G now transforms this integral. Since G is compact and has no boundary, the divergence theorem gives $\int_G \mathrm{div}(X) \, d\mu = 0$ for every smooth vector field X. Applying this to $X = f \, \nabla f$ and expanding using the product rule $\mathrm{div}(f \, \nabla f) = |\nabla f|^2 + f \, \Delta f$ yields

$$\int_G |\nabla f|^2 \, d\mu \;+\; \int_G f \, \Delta f \, d\mu \;=\; 0,$$

and therefore

$$\int_G f \, \Delta f \, d\mu \;=\; -\int_G |\nabla f|^2 \, d\mu.$$

Together with the equation above, this gives

$$0 \;=\; -\int_G |\nabla f|^2 \, d\mu.$$

Since $|\nabla f|^2 \geq 0$ pointwise and its integral is zero, the gradient vanishes everywhere on G. A smooth function with everywhere-zero gradient on a connected manifold is constant. Any two points of G

can be joined by a smooth path, and along that path the derivative of f equals ∇f dotted with the tangent vector, which is zero, so f takes the same value at both endpoints.

Hence $\ker \Delta = \operatorname{span}\{\mathbf{1}\}$, which corresponds exactly to $\mathcal{H}_{\mathbf{1}}$, the Peter–Weyl block of the trivial representation $\mathbf{1}$. For every non-trivial π, the functions in $\mathcal{H}_\pi$ are orthogonal to constants (by the Peter–Weyl theorem), so they are non-constant and hence not harmonic, giving $\lambda_\pi > 0$. (Compactness is essential: on a non-compact group the kernel of Δ can be larger.) □

6 Convergence to Haar Measure

We now use the Peter–Weyl spectral decomposition to prove the main convergence theorem: the law of Brownian motion on G converges weakly to the Haar measure μ as $t \to \infty$.

The convergence is the spectral consequence of the Peter–Weyl decomposition, and the intuition is the same as on the circle. On $U(1)$, the heat semigroup sends $e^{ik\theta}$ to $e^{-k^2t/2}e^{ik\theta}$, so every nonzero Fourier mode decays and only the constant mode survives. On G the same thing happens mode by mode. Constant functions belong to the trivial representation $\mathbf{1}$ and satisfy $\lambda_{\mathbf{1}} = 0$, so the heat semigroup leaves them unchanged. Every nontrivial block $\mathcal{H}_\pi$ carries a strictly negative eigenvalue $-\lambda_\pi < 0$ of Δ, so the heat semigroup damps functions in $\mathcal{H}_\pi$ by the factor $e^{-\lambda_\pi t/2}$, which tends to zero as $t \to \infty$. In the limit, all nontrivial components are wiped out and only the projection onto the constants survives, which is exactly the average of the function against Haar measure.

6.1 Heat Semigroup on Eigenspaces

Recall from Section 4 that the generator of Brownian motion is $\mathcal{A} = \frac{1}{2}\Delta$. By Corollary 5.8, each $\mathcal{H}_\pi$ is an eigenspace of Δ with eigenvalue $-\lambda_\pi$, so

$$\mathcal{A}f \;=\; -\frac{\lambda_\pi}{2}\, f \qquad \text{for all } f \in \mathcal{H}_\pi.$$

The heat equation $\partial_t u = \mathcal{A}u$ restricted to the mode $\mathcal{H}_\pi$ is

$$\partial_t u_\pi \;=\; -\frac{\lambda_\pi}{2}\, u_\pi, \qquad u_\pi(0) = f.$$

This ODE has the explicit solution $u_\pi(t) = e^{-\lambda_\pi t/2} f$. Recalling that $P_t f$ solves the heat equation with initial datum f, we obtain:

$$P_t f \;=\; e^{-\lambda_\pi t/2}\, f \qquad \text{for all } f \in \mathcal{H}_\pi.$$

6.2 Semigroup Expansion

Proposition 6.1. *For any $f \in L^2(G,\mu)$,*

$$P_t f \;=\; c\cdot \mathbf{1} \;+\; \sum_{\pi\neq\mathbf{1}} \sum_{i,j=1}^{d_\pi} e^{-\lambda_\pi t/2}\, a_{ij}^{\pi}\, \pi_{ij},$$

where $c = \langle f, \mathbf{1}\rangle_{L^2(G)} = \int_G f\, d\mu$ and a_{ij}^{π} are the Peter–Weyl coefficients of f.

Proof. By the Peter–Weyl theorem, f expands as

$$f \;=\; c\cdot\mathbf{1} \;+\; \sum_{\pi\neq\mathbf{1}} \sum_{i,j=1}^{d_\pi} a_{ij}^{\pi}\, \pi_{ij}, \qquad c = \int_G f\, d\mu.$$

Applying P_t and using $P_t\pi_{ij} = e^{-\lambda_\pi t/2}\pi_{ij}$ (which follows from $\pi_{ij} \in \mathcal{H}_\pi$ and $\lambda_{\mathbf{1}} = 0$) gives the result. □

Since $\lambda_\pi > 0$ for every $\pi \neq \mathbf{1}$ (Proposition 5.9), every non-trivial term in the expansion decays exponentially. Hence

$$P_t f \;\longrightarrow\; c\cdot\mathbf{1} \;=\; \Big(\int_G f\, d\mu\Big)\mathbf{1} \qquad \text{as } t\to\infty.$$

6.3 Convergence of the Law

Theorem 6.2 (Convergence to Haar measure)**.** *Let $(U_t)_{t\geq 0}$ be the Brownian motion on G started at the identity e, and let ν_t denote its law. Then $\nu_t \Rightarrow \mu$ weakly as $t\to\infty$.*

Proof. Recall that $P_t f(g) = \mathbb{E}[f(U_t) \mid U_0 = g]$, so evaluating at $g = e$ gives

$$P_t f(e) \;=\; \mathbb{E}[f(U_t)] \;=\; \int_G f\, d\nu_t.$$

It therefore suffices to show that $P_t f(e) \to \int_G f\, d\mu$ for every $f \in C(G)$. We do this in three steps.

Step 1: Convergence for Peter–Weyl polynomials. A Peter–Weyl polynomial is a finite linear combination $q = c\cdot\mathbf{1} + \sum_{\text{finite}} a_{ij}^{\pi}\, \pi_{ij}$. Since the sum is finite, we can apply P_t termwise using $P_t\pi_{ij} = e^{-\lambda_\pi t/2}\pi_{ij}$ (from Section 5), giving

$$P_t q \;=\; c\cdot\mathbf{1} \;+\; \sum_{\text{finite}} e^{-\lambda_\pi t/2}\, a_{ij}^{\pi}\, \pi_{ij}.$$

Each $\lambda_\pi > 0$ for $\pi \neq \mathbf{1}$ (Proposition 5.9), so every nontrivial term decays to zero as $t\to\infty$. Because the sum is finite, the decay is uniform in g, and we obtain $P_t q(g) \to c = \int_G q\, d\mu$ uniformly over all $g \in G$.

Step 2: Density. By the Peter–Weyl theorem (Theorem 5.2), Peter–Weyl polynomials are uniformly dense in $C(G)$. So for any $f \in C(G)$ and any $\varepsilon > 0$, there exists a Peter–Weyl polynomial q with $\|f - q\|_\infty < \varepsilon$.

Step 3: Approximation via the Markov property. Since $P_t f(g) = \mathbb{E}[f(U_t) \mid U_0 = g]$, the operator P_t is a Markov operator. In particular, for any two bounded continuous functions f and q,

$$|P_t f(g) - P_t q(g)| \;\le\; \mathbb{E}\big[|f(U_t) - q(U_t)| \;\big|\; U_0 = g\big] \;\le\; \|f - q\|_\infty.$$

The first inequality is the triangle inequality inside the expectation, and the second bounds $|f - q|$ by its supremum. Choose q as in Step 2. Then for any $t \ge 0$,

$$\begin{aligned} \left|P_t f(e) - \textstyle\int_G f\, d\mu\right| \;&\le\; |P_t f(e) - P_t q(e)| + \left|P_t q(e) - \textstyle\int_G q\, d\mu\right| + \left|\textstyle\int_G q\, d\mu - \int_G f\, d\mu\right| \\ &\le\; \|f - q\|_\infty + \left|P_t q(e) - \textstyle\int_G q\, d\mu\right| + \|f - q\|_\infty. \end{aligned}$$

The outer two terms are each less than ε. By Step 1, the middle term tends to zero as $t \to \infty$. So for all large enough t, the entire expression is less than 3ε. Since $\varepsilon > 0$ was arbitrary, $P_t f(e) \to \int_G f\, d\mu$.

Combined with $P_t f(e) = \int_G f\, d\nu_t$, this gives $\int_G f\, d\nu_t \to \int_G f\, d\mu$ for every $f \in C(G)$, which is precisely weak convergence $\nu_t \Rightarrow \mu$. □

7 Conclusion

This paper has constructed and analyzed Brownian motion on the unitary group $\mathrm{U}(n)$, tracing a path that runs through three distinct areas of mathematics before arriving at a single clean result.

What was built. Starting from the defining constraint $U^*U = I$, we identified the Lie algebra $\mathfrak{u}(n)$ of skew-Hermitian matrices as the space of infinitesimal motions, equipped $\mathrm{U}(n)$ with the Hilbert–Schmidt bi-invariant metric, and derived the Laplace–Beltrami operator $\Delta = \sum_k R_k^2$ from the geometry. Brownian motion on $\mathrm{U}(n)$ was then characterized in four equivalent ways: as the solution to the martingale problem for $\frac{1}{2}\Delta$, as the Markov process governed by the heat semigroup $P_t = e^{t\frac{1}{2}\Delta}$, via its transition density (the heat kernel), and as the explicit solution of the Stratonovich SDE

$$dU_t \;=\; \sum_{k=1}^{n^2} X_k U_t \circ dB_t^k.$$

The equivalence between the Stratonovich and Itô formulations illuminated why the correction drift $-\frac{1}{2}U_t\, dt$ is not an artifact but is precisely what keeps the process on $\mathrm{U}(n)$: it cancels the quadratic-variation term that would otherwise push U_t off the group.

The role of compactness. Compactness of $\mathrm{U}(n)$ was the decisive structural feature throughout. It guaranteed the existence and uniqueness of Haar measure, ensured ellipticity of the generator (so the martingale problem is well-posed), and forced every harmonic function to be constant (giving $\ker \Delta_n = \operatorname{span}\{\mathbf{1}\}$). Most importantly, compactness ensured that each nontrivial Peter–Weyl mode $\mathcal{H}_\pi$ carries a strictly negative eigenvalue $-\lambda_\pi < 0$, so the corresponding component of the heat semigroup decays as $e^{-\lambda_\pi t/2}$ and the process converges to equilibrium.

On a noncompact group, none of these conclusions hold in general. Haar measure is typically infinite and cannot be normalized to a probability measure, so there is no candidate limiting distribution. Harmonic functions need not be constant, meaning the kernel of Δ can be larger than the span of the constant function. And without a guarantee that nontrivial modes carry strictly negative eigenvalues, exponential convergence to a uniform equilibrium is not automatic.

Peter–Weyl as the engine of convergence. The Peter–Weyl theorem played the role that classical Fourier analysis plays on the circle. It decomposed $L^2(G, \mu)$ into finite-dimensional blocks $\mathcal{H}_\pi$, each of which is an eigenspace of Δ with eigenvalue $-\lambda_\pi$. The key algebraic fact, that Δ_n is scalar on each block, followed from Schur's lemma applied to the $G \times G$ action, where the bi-invariance of the metric was again essential. With this spectral picture in hand, the convergence result reduced to a direct calculation: every non-constant Fourier mode decays as $e^{-\lambda_\pi t/2}$, leaving only the constant mode (the integral against μ) in the limit.

The main result. The convergence theorem (Theorem 6.2) states that the law ν_t of $(U_t)_{t \geq 0}$, started at the identity, satisfies $\nu_t \Rightarrow \mu$ weakly as $t \to \infty$. In probabilistic terms: the process eventually forgets its initial position and spreads uniformly over $\mathrm{U}(n)$ according to the Haar measure.

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